EXTENSION 1

Stage 6

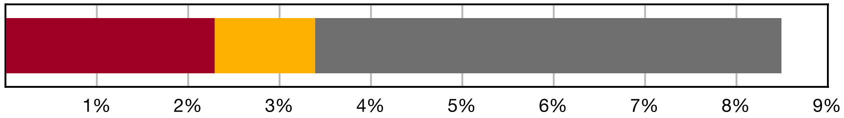
Statistical Analysis (Ext1), S1 The Binomial (Y12)

Statistics and Binomial Distributions (Ext1)

Teacher: SHS Maths

Exam Equivalent Time: 36 minutes *(based on allocation of 1.5 minutes per mark)*

S1 The Binomial



Binomial Probability
Statistics and Binomial Distributions
Normal Approx and Sample Proportions

***SmarterMaths analytics based on the average contribution to Extension 1 exams since 2020.**

HISTORICAL CONTRIBUTION

- *S1 The Binomial* has contributed an average of 8.6% per new syllabus exam since it was was introduced in 2020.
- This topic has been split into three categories for analysis purposes which are: *1-Binomial Probability (2.3%), 2-Statistics and Binomial Distributions (1.2%) and 3-Normal Approximations of Sample Proportions (5.1%).*
- This analysis looks at *Statistics and Binomial Distributions.*

HSC ANALYSIS - What to expect and common pitfalls

- *Statistics and Binomial Distributions* involves new HSC content that looks at the mean and variance of two-outcome (Bernoulli) distributions.
- Most recently examined in 2020 Ext1 exams in a low difficulty 3-mark question that incorporated normal distribution.
- The question database is informed by specific syllabus dot points, NESA Topic Guidance and the exam history of other States that cover similar content.
- We note that Topic Guidance exemplar questions have shown this topic is a natural cross-topic fit with *Binomial Probability and* recommend a review of *S1 EQ-Bank 13*.
- Further to the above point, students must be able to readily identify binomial distributions from two-outcome scenarios and apply statistical analysis to them. This has proven challenging in other States and numerous examples in the database require a deep understanding of this concept.

Questions

1. Statistics, EXT1 S1 EQ-Bank 1 MC

If X equals the number of successes in n independent Bernoulli trials, how many distinct values can X take?

A. $n-1$

B. $n(n-1)$

C. n

D. $n+1$

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2. Statistics, EXT1 S1 2008 MET2 5 MC

Let X be a discrete random variable with a binomial distribution. The mean of X is 1.2 and the variance of X is 0.72

The values of n (the number of independent trials) and p (the probability of success in each trial) are

A. $n=3, \quad p=0.6$

B. $n=2, \quad p=0.6$

C. $n=2, \quad p=0.4$

D. $n=3, \quad p=0.4$

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3. Statistics, EXT1 S1 SM-Bank 4 MC

When a standard 6-sided die is thrown, the probability that it shows a prime number is $\dfrac{2}{3}$.

If 10 standard dice are thrown, the number, N, of times a prime number is showing has a binomial distribution.

What is the standard deviation of N, correct to 3 decimal places?

A. 0.222

B. 0.471

C. 1.491

D. 2.222

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4. Statistics, EXT1 S1 2020 HSC 12b

When a particular biased coin is tossed, the probability of obtaining a head is $\dfrac{3}{5}$.

This coin is tossed 100 times.

Let X be the random variable representing the number of heads obtained. This random variable will have a binomial distribution.

i. Find the expected value, $E(X)$. ***(1 mark)***

ii. By finding the variance, $\text{Var}(X)$, show that the standard deviation of X is approximately 5. ***(1 mark)***

iii. By using a normal approximation, find the approximate probability that X is between 55 and 65. ***(1 mark)***

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5. Statistics, EXT1 S1 EQ-Bank 10

Four cards are placed face down on a table. The cards are made up of a Jack, Queen, King and Ace.

A gambler bets that she will choose the Queen in a random pick of one of the cards.

If this process is repeated 7 times, express the gambler's success as a Bernoulli random variable and calculate

i. the mean. ***(1 mark)***

ii. the variance. ***(1 mark)***

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6. Statistics, EXT1 S1 EQ-Bank 13

On average, batsmen playing cricket in a T20 competition play a scoring shot two out of every three balls.

In a regular season, a total of 1200 overs that each contain six balls, are bowled.

Estimate how many overs would have at least five scoring shots. ***(3 marks)***

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7. Statistics, EXT1 S1 SM-Bank 4

In an experiment, a pair of dice are rolled 70 times.

A success is recorded if the sum of the dice roll is 5 or less.

i. What is the mean of this binomial distribution?
Give your answer to one decimal place. ***(3 marks)***

ii. What is the standard deviation?
Give your answer to one decimal place. ***(1 mark)***

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8. Statistics, EXT1 S1 2015 MET2 10

The binomial random variable, X, has $E(X)=2$ and $\text{Var}(X)=\dfrac{4}{3}$.

Calculate $P(X=1)$. ***(3 marks)***

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9. Statistics, EXT1 S1 2017 MET2 18

Let X be a discrete random variable with binomial distribution $X \sim \text{Bin}(n,p)$. The mean and the standard deviation of this distribution are equal.

Given that $0<p<1$, what is the smallest number of trials, n, such that $p \le 0.01$. ***(2 marks)***

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10. Statistics, EXT1 S1 2012 MET2 3

Steve and Jess are two students who have agreed to take part in a psychology experiment. Each has to answer several sets of multiple-choice questions. Each set has the same number of questions, n, where n is a number greater than 20. For each question there are four possible options A, B, C or D, of which only one is correct.

a. Steve decides to guess the answer to every question, so that for each question he chooses A, B, C or D at random.

Let the random variable X be the number of questions that Steve answers correctly in a particular set.

i. What is the probability that Steve will answer the first three questions of this set correctly? ***(1 mark)***

ii. Use the fact that the variance of X is $\dfrac{75}{16}$ to show that the value of n is 25. ***(1 mark)***

b. The probability that Jess will answer any question correctly, independently of her answer to any other question, is $p\ (p>0)$. Let the random variable Y be the number of questions that Jess answers correctly in any set of 25.

If $P(Y>23)=6\times P(Y=25)$, show that the value of $p=\dfrac{5}{6}$. ***(2 marks)***

Worked Solutions

1. Statistics, EXT1 S1 EQ-Bank 1 MC

Distinct values of X in:

$1\text{ trial} = 2 \quad (X=0 \text{ or } 1)$

$2\text{ trials} = 3 \quad (X=0, 1 \text{ or } 2)$

$\vdots$

$n\text{ trials} = n+1 \quad (X=0,1,\ldots,n)$

$\Rightarrow \ D$

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2. Statistics, EXT1 S1 2008 MET2 5 MC

$X \sim \text{Bin}(n,p)$

$\mu = 1.2$

$np = 1.2 \ldots (1)$

$\text{Var}(X) = 0.72$

$np(1-p) = 0.72 \ldots (2)$

Solve simultaneous equations:

$\therefore n=3, \quad p=0.4$

$\Rightarrow \ D$

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3. Statistics, EXT1 S1 SM-Bank 4 MC

$$N \sim \text{Bin}(n,p) \sim \text{Bin}\left(10, \frac{2}{3}\right)$$

$$\begin{aligned}\text{Var}(N) &= np(1-p)\\ &= 10\cdot\frac{2}{3}\left(1-\frac{2}{3}\right)\\ &= \frac{20}{9}\end{aligned}$$

$$\begin{aligned}\therefore \sigma_N &= \sqrt{\frac{20}{9}}\\ &= 1.4907\ldots\end{aligned}$$

$\Rightarrow \ C$

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4. Statistics, EXT1 S1 2020 HSC 12b

i. $X = \text{number of heads}$

$$X \sim \text{Bin}(n,p) \sim \text{Bin}\left(100, \frac{3}{5}\right)$$

$$\begin{aligned}E(X) &= np\\ &= 100\times\frac{3}{5}\\ &= 60\end{aligned}$$

ii. $$\begin{aligned}\text{Var}(X) &= np(1-p)\\ &= 60\times\frac{2}{5}\\ &= 24\end{aligned}$$

$$\begin{aligned}\sigma(x) &= \sqrt{24}\\ &\approx 5\end{aligned}$$

iii. $$\begin{aligned}P(55\le x\le 65) &\approx P(-1\le z\le 1)\\ &\approx 68\%\end{aligned}$$

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5. Statistics, EXT1 S1 EQ-Bank 10

i. Let $X = \text{number of Queens chosen}$

$$X \sim \text{Bin}\left(7, \frac{1}{4}\right)$$

$$\begin{aligned}E(X) &= np\\ &= 7\times\frac{1}{4}\\ &= \frac{7}{4}\end{aligned}$$

ii. $$\begin{aligned}\text{Var}(X) &= np(1-p)\\ &= \frac{7}{4}\left(1-\frac{1}{4}\right)\\ &= \frac{21}{16}\end{aligned}$$

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6. Statistics, EXT1 S1 EQ-Bank 13

Let $X =$ number of scoring shots in a six ball over

$$X \sim \text{Bin}\left(6, \frac{2}{3}\right)$$

$$\begin{aligned} P(X \geq 5) &= P(X=5) + P(X=6) \\ &= {}^6C_5 \cdot \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right) + {}^6C_6 \cdot \left(\frac{2}{3}\right)^6 \\ &= \frac{64}{243} + \frac{64}{729} \\ &= \frac{256}{729} \end{aligned}$$

Let $Y =$ number of overs with at least 5 scoring shots

$$Y \sim \text{Bin}\left(1200, \frac{256}{729}\right)$$

$$\begin{aligned} E(Y) &= np \\ &= 1200 \times \frac{256}{729} \\ &= 421.39\ldots \\ &= 421 \quad \text{(nearest over)} \end{aligned}$$

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7. Statistics, EXT1 S1 SM-Bank 4

i. Array of possible roll totals:

	1	2	3	4	5	6
1	2	3	4	5		
2	3	4	5			
3	4	5				
4	5					
5						
6						

STRATEGY: A table (or array) can be a very efficient and error minimising strategy in questions like this.

$$P(5 \text{ or less}) = \frac{10}{36} = \frac{5}{18}$$

Let $X =$ number of rolls ≤ 5

$$X \sim \text{Bin}\left(70, \frac{5}{18}\right)$$

$$\begin{aligned} E(X) &= np \\ &= 70 \times \frac{5}{18} \\ &= 19.4 \end{aligned}$$

ii. $\text{Var}(X) = np(1-p)$

$$\begin{aligned} \sigma^2 &= 70 \times \frac{5}{18}\left(1-\frac{5}{18}\right) \\ &= 14.043 \\ \therefore \sigma &= 3.7 \quad \text{(to 1 d.p.)} \end{aligned}$$

8. Statistics, EXT1 S1 2015 MET2 10

$np = 2 \ldots (1)$

$np(1-p) = \dfrac{4}{3} \ldots (2)$

Solve simultaneous equations:

Substitute $np = 2$ into (2)

$$\begin{aligned} 2(1-p) &= \frac{4}{3} \\ 2 - 2p &= \frac{4}{3} \\ p &= \frac{1}{3} \end{aligned}$$

$n = 6,\quad p = \dfrac{1}{3}$

$\therefore X \sim \text{Bin}\left(6, \dfrac{1}{3}\right)$

$$\begin{aligned} \therefore P(X=1) &= \binom{6}{1} \times \left(\frac{1}{3}\right)^1 \times \left(\frac{2}{3}\right)^5 \\ &= 6 \times \frac{1}{3} \times \left(\frac{2}{3}\right)^5 \\ &= \frac{64}{243} \end{aligned}$$

9. Statistics, EXT1 S1 2017 MET2 18

$\mu = np,\quad \text{Var}(X) = np(1-p) = \sigma_x^2$

Given $\mu = \sigma_x$,

$$\begin{aligned} np &= \sqrt{np(1-p)} \\ n^2p^2 &= np(1-p) \\ np(np - 1 + p) &= 0 \\ np - 1 + p &= 0, \quad (np \neq 0) \\ p(n+1) &= 1 \\ p &= \frac{1}{n+1} \end{aligned}$$

$$\begin{aligned} \frac{1}{n+1} &\leq \frac{1}{100} \\ n + 1 &\geq 100 \\ n &\geq 99 \end{aligned}$$

$\therefore n_{\text{min}} = 99$

10. Statistics, EXT1 S1 2012 MET2 3

a.i. $P(3\text{ correct in a row}) = \left(\dfrac{1}{4}\right)^3$

$$= \frac{1}{64}$$

a.ii. $\text{Var}(X) = np(1-p)$

$$\frac{75}{16} = n\left(\frac{1}{4}\right)\left(\frac{3}{4}\right)$$

$$75 = 3n$$

$$\therefore n = 25$$

b. $Y \sim \text{ Bin}(25, p)$

$$\begin{aligned} P(Y > 23) &= 6 \times P(Y = 25) \\ P(Y = 24) + P(Y = 25) &= 6 \times P(Y = 25) \\ P(Y = 24) &= 5 \times P(Y = 25) \\ \binom{25}{24} p^{24}(1-p)^1 &= 5p^{25} \\ 25p^{24}(1-p) &= 5p^{25} \\ 25p^{24} - 25p^{25} - 5p^{25} &= 0 \\ 25p^{24} - 30p^{25} &= 0 \\ 5p^{24}(5-6p) &= 0 \end{aligned}$$

♦♦♦ *Mean mark part (c) 19%.*

$$\therefore p = \frac{5}{6}, \quad (p > 0) \quad \ldots \text{as required}$$